Lidar Introduction and Use

1. Course Content

- Learn about the hardware specifications of the lidar sensor equipped on the A1 Pro.
- Launch the lidar driver and view real-time lidar scan point cloud in RViz2.
- Use ROS2 command-line tools to check the radar data format of the `/scan` topic.

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2. Lidar Parameters Overview

The A1 Pro robot supports **T-mini Plus** and **Slam C1** lidars:

2.1 T-mini Plus Lidar Parameters

Attribute	Value	Unit
Ranging Frequency	4000	Hz
Scan Frequency	6 (6~12)	Hz
Ranging Range	0.05 ~ 12	m
Scan Angle	0 ~ 240	Deg
Ranging Accuracy	20	mm
Angle Resolution	0.54	Deg

2.2 Slam C1

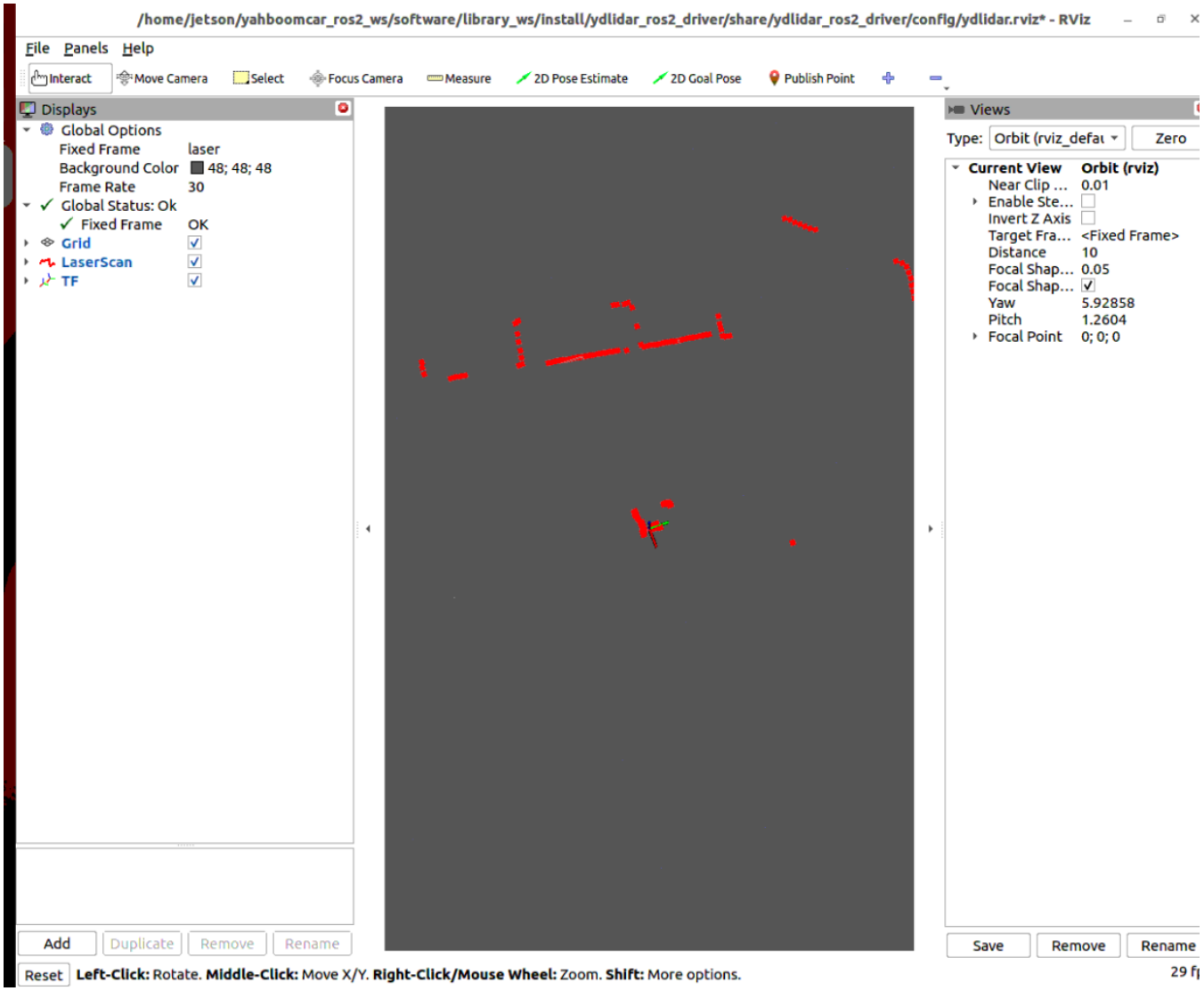
Attribute	Value	Unit
Ranging Frequency	5000	Hz
Scan Frequency	8 (8-12)	Hz
Ranging Range	0.05-12	m
Scan Angle	0-240	Deg
Ranging Accuracy	30	mm
Angle Resolution	0.72	Deg

3. Demo Examples

3.1 Launch Lidar Driver and RViz Visualization

Run in terminal:

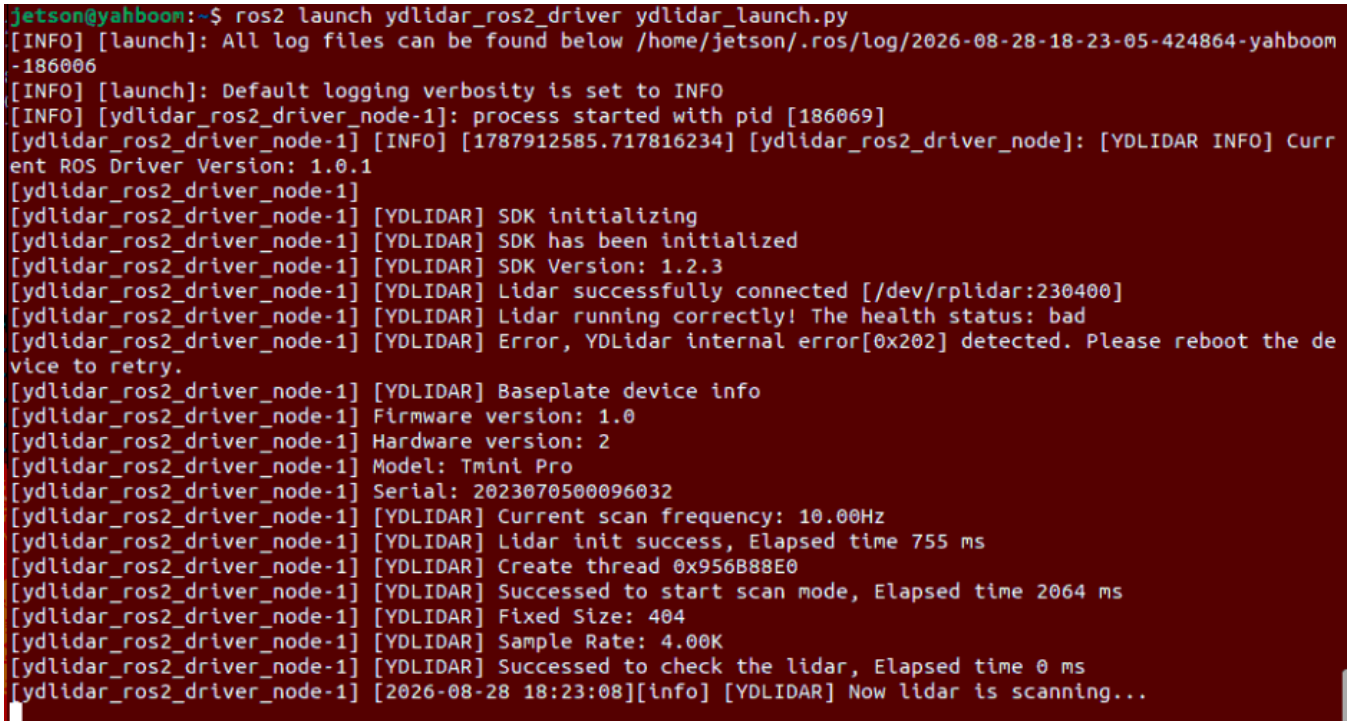
```
# Launch TminiPlus lidar and open RViz visualization
ros2 launch ydyardar_ros2_driver ydlidar_launch_view.py
# Launch C1 lidar and open RViz visualization data
ros2 launch sllidar_ros2 view_sllidar_c1_launch.py
```



In RViz2, the red point cloud represents the lidar's real-time scan of the surrounding environment. In the **LaserScan** plugin panel on the left side of RViz, you can adjust the point cloud display size (Size), color style (Color Transformer), and decay time (Decay Time).

> **Tip**
>
> If you only need to start the lidar node in the background (without automatically opening RViz), run:

```
# Start Tminiplus lidar
ros2 launch ydlidar_ros2_driver ydlidar_launch.py
# Start C1 lidar
ros2 launch sllidar_ros2 sllidar_c1_launch.py
```



4. Viewing Lidar Data

Run ROS2 CLI command in a new terminal to view raw lidar data:

```
ros2 topic echo /scan
```

```
jetson@yahboom:~$ ros2 topic echo /scan
header:
  stamp:
    sec: 1787912653
    nanosec: 767115000
  frame_id: laser
angle_min: -3.1415927410125732
angle_max: 3.1415927410125732
angle_increment: 0.015591030940413475
time_increment: 0.0002514962397981435
scan_time: 0.10034699738025665
range_min: 0.029999999329447746
range_max: 12.0
ranges:
- 0.0
- 2.1600000858306885
- 2.1549999713897705
- 2.1500000953674316
- 2.1459999084472656
- 2.1429998874664307
- 2.1410000324249268
- 2.1440000534057617
- 2.1440000534057617
- 2.1449999809265137
- 2.1489999294281006
- 2.1540000438690186
- 2.1579999923706055
- 2.1600000858306885
- 2.1659998893737793
- 2.1679999828338623
- 2.1700000762939453
- 2.177999973297119
- 2.183000087738037
- 2.140000104904175
- 1.9279999732971191
- 1.809999942779541
- 1.7380000352859497
- 1.7369999885559082
- 1.74399995803833
- 1.75600004196167
- 1.7640000581741333
- 1.7680000066757202
- 1.781999945640564
- 1.7910000085830688
```

4.1 Core Data Field Parsing

- `header.frame_id`: Lidar coordinate frame name (usually `laser` or `laser_frame`).
- `angle_min` / `angle_max`: Starting and ending radians of lidar scan (0° ~ 360° corresponds to `0` to `2π`).
- `angle_increment`: Angular resolution increment between adjacent points (radians).
- `range_min` / `range_max`: Shortest and longest valid ranging distances of the lidar (meters).
- `ranges`: Array of measured distances at each angle within the current scan frame (meters).